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1 Product Requirements Document (PRD): Tesla Optimus Gen 3 Humanoid Robot

1.1 1. Overview

1.1.1 1.1 Purpose

This document outlines the product requirements for **Tesla Optimus Gen 3**, a humanoid robot intended for industrial and future consumer applications. The Gen 3 model introduces a 22-degree-of-freedom (DoF) hand and tendon-driven architecture, aiming to achieve human-level dexterity while remaining cost-effective for mass production. The PRD serves to align design teams, engineering, manufacturing, and business stakeholders on goals, constraints, functionality, and milestones.

1.1.2 1.2 Background and Context

Tesla first announced the Optimus robot at AI Day 2021. Prototypes were demonstrated at AI Day 2022 (Gen 1) and December 2023 (Gen 2). Gen 2 introduced 11-DoF

hands with tactile sensors and improved mobility. Elon Musk’s vision is to deploy Optimus robots in Tesla factories and later in homes, with a long-term goal of producing millions by 2029.

Gen 3 is positioned as the first design intended for mass production. Tesla plans to repurpose its Fremont factory for Optimus and invest over \$20 billion in 2026 toward robot and autonomous vehicle manufacturing. Production will initially be slow due to new parts and processes.

1.2 2. Problem Statement and Goals

1.2.1 2.1 Problem Statement

Humanoid robots currently suffer from limited dexterity, short battery life, high cost, and dependency on teleoperation. Industrial use cases are constrained by low reliability and complex supply chains. Tesla aims to deliver a general-purpose humanoid robot that can perform tasks in unstructured environments, starting with structured factory operations.

1.2.2 2.2 Goals and Objectives

1. **Dexterous Manipulation:** Achieve near-human hand dexterity with 22 DoF hands and tendon-driven design.
2. **Mass Production:** Create a design suitable for high-volume manufacturing. Eliminate weight and complexity by relocating finger actuators to the forearm.
3. **Cost Reduction:** Target a long-term selling price of \$20k-\$30k per robot; reduce the current bill of materials (BOM), where hands account for ~17 % of cost (~\$9,500 per unit).
4. **Internal Deployment:** Deploy thousands of robots in Tesla factories by 2026-2027 to automate repetitive tasks like parts handling, assembly, and logistics.
5. **Prepare for Consumer Market:** Lay the groundwork for future household tasks (folding laundry, cooking, cleaning) and potential service roles.

1.3 3. Product Description and Scope

1.3.1 3.1 System Overview

- **Hardware:** Humanoid platform with 2 arms, 2 legs, torso, head, and two 22-DoF hands. Each hand has five fingers with four joints per finger plus two wrist joints. Tendon-driven architecture moves actuators to the forearm, reducing hand weight and size. Lightweight metal skeleton with steel cable tendons and springs. Integrated cameras and force sensors provide tactile and visual feedback.
- **Software:** A single neural network (“Foundation Model”) handles locomotion, manipulation, perception, and decision-making. Robot learns from human demonstrations and simulation training. Vision and sensor data feed into Tesla’s Full Self-Driving AI stack for environment understanding.

- **Power:** Battery capacity targets 8–20 hours of operation, though current prototypes run 2–4 hours. Batteries reside in torso; weight trade-offs will be evaluated.
- **Safety:** Tendon-driven compliance, force sensors, and clutches protect humans. Failsafe shutdown prevents harmful movement. Robot adheres to emerging safety standards for industrial mobile robots.

1.3.2 3.2 In-Scope Functional Requirements

1. Hand Dexterity

- Perform precision grips (e.g., pinch, hook, spherical grasp) and power grasps.
- Manipulate small objects (e.g., screws, electronic components) and fragile items (e.g., egg).
- Support multi-finger in-hand manipulation (e.g., turning a knob, threading a wire).

2. Locomotion and Balance

- Walk forward at up to 5 mph, traverse uneven terrain, climb stairs.
- Maintain dynamic balance under external disturbances; recover from slips.

3. Factory Automation Tasks

- Pick and place parts and assemblies.
- Carry loads up to 45 lbs while walking.
- Operate simple tools (screwdrivers, wrenches) under supervision.
- Assist with material logistics: move bins, pallets, components.

4. Service Tasks (Future Scope)

- Perform household chores (fold laundry, vacuum, cook).
- Provide human interaction (greeting, delivering items, simple conversation).

5. User Interaction

- Screen interface on head displays status and basic communication; voice command support planned.
- Safety interface: emergency stop accessible to humans.

1.3.3 3.3 Out-of-Scope

- Advanced autonomous decision-making in unstructured, unpredictable environments (e.g., disaster zones).
- High-speed running or athletic movements (beyond walking and stair climbing).
- Teleoperation beyond initial testing; goal is limited human intervention.

1.4 4. Design and Engineering Requirements

1.4.1 4.1 Mechanical Requirements

- **Hand Mechanism:** 22 DoF using tendon cables and linear actuators in forearm; each tendon must be rated for repeated cycles and maintain tension via springs.

Joints should include passive compliance to absorb impact.

- **Materials:** Use lightweight aluminum/magnesium alloy for bones; high-strength steel or braided synthetic fiber for tendons. Plastic covers for aesthetics and protection.
- **Weight Limits:** Target overall mass < 125 lbs to preserve mobility. Hand weight < 1.5 lbs each.
- **Force Output:** Achieve grip force sufficient to lift objects up to 45 lbs; peak finger force 10–15 N.

1.4.2 4.2 Electrical and Electronics Requirements

- **Actuators:** Custom servo motors with $\sim 0.05^\circ$ position accuracy and 3 ms response time (based on analyses) to enable precise finger control.
- **Sensors:**
 - Tactile sensors in fingers (force/pressure) for closed-loop grip control.
 - Cameras in head and potentially hands for vision; wide field-of-view to perceive environment.
 - Inertial measurement units (IMUs) in torso and limbs for balance.
 - Torque sensors at joints for force estimation.
- **Battery:** Modular pack enabling hot-swapping; design must support 2–4 hour operation initially with roadmap to 8–12 hours via higher-density cells.

1.4.3 4.3 Software Requirements

- **Control Architecture:** Unified neural network controlling locomotion, manipulation, and perception. System should support real-time inference on custom D1 chips.
- **Learning & Simulation:** Leverage simulation acceleration to train thousands of virtual robots; incorporate reinforcement learning and imitation learning for manipulation tasks.
- **Safety & Overrides:** Real-time monitoring of joint torques, speeds, and temperatures; override commands for emergency stop; fallback modes if sensors fail.

1.4.4 4.4 Manufacturing and Supply Chain

- **Suppliers:** Use established Chinese suppliers for actuators, motors, harmonic reducers, bearings, sensors, and batteries; key partners include Sanhua Intelligent Control, Green Harmonic, Tuopu Group, and Keli Sensing. Final assembly occurs in U.S. plants.
- **Component Standardization:** Design actuators and tendons for mass production with minimal variation. Ensure compatibility with existing Tesla manufacturing processes.
- **Quality Assurance:** Implement rigorous component testing for durability, including > 100,000 cycles on tendon units; incorporate statistical process control.

1.5 5. User Stories and Use Cases

1. **Factory Operator:** As a manufacturing engineer, I want Optimus robots to load and unload automotive parts so that human workers can focus on higher-level assembly tasks.
2. **Logistics Supervisor:** As a logistics manager, I need robots to transport parts across the warehouse without human intervention, reducing manual cart traffic.
3. **Quality Inspector:** As an inspector, I want Optimus to pick up car components, present them under cameras, and gently rotate them for visual inspection.
4. **Homeowner (Future):** As a homeowner, I want the robot to fold laundry and vacuum floors so that I save time on household chores.
5. **Researcher:** As an AI developer, I need a consistent API to program new manipulation tasks and collect training data for continuous improvement.

1.6 6. Success Metrics and KPIs

- **Dexterity:** Ability to achieve > 90 % success rate on pick-and-place tasks of varied sizes; ability to handle delicate objects without damage.
- **Production Scale:** Number of robots produced per month. Target 5,000 units in 2026 progressing toward 50,000+ annually by 2027-2028.
- **Cost per Unit:** Reduce BOM to <\$30k by 2027; hands cost < \$4k (currently ~\$9.5k).
- **Runtime:** Achieve $\geq$ 8-hour battery life within three development iterations.
- **Task Completion:** 95 % completion rate on targeted factory tasks without teleoperation; user satisfaction for household tasks in pilot programs.

1.7 7. Dependencies and Risks

1.7.1 7.1 Dependencies

- Availability of high-precision actuators and harmonic reducers from suppliers.
- Continued AI development and improvements in simulation platforms.
- Battery technology improvements for higher energy density.
- Regulatory approval and safety standards (IEEE/ISO) for humanoid robots.

1.7.2 7.2 Risks

- **Technical Complexity:** Achieving human-level dexterity may prove harder than anticipated. Tendon-driven systems suffer from friction and compliance that complicate control.
- **Supply Chain:** Geopolitical tensions and export controls could disrupt supply of rare-earth materials and critical components.
- **Production Delays:** Past targets slipped; only hundreds of robots were produced by mid-2025. Initial production of Gen 3 will be “agonizingly slow”.
- **Cost Overruns:** High BOM costs, especially in hands and actuators, may delay achieving the \$20k-\$30k price point.
- **Safety:** Insufficient compliance or control could lead to accidents; new safety standards are still evolving.

1.8 8. Milestones and Timeline

Quarter	Milestone	Description
Q1 2026	Gen 3 Design Finalization	Unveil final 22-DoF hand and tendon-driven architecture; commence tooling for mass production.
Q2 2026	Pilot Production at Fremont	Begin low-volume production using retooled Fremont factory; integrate supply chain components from Chinese partners.
Q3 2026	Internal Deployment	Deploy 50–100 robots in Tesla factories for parts handling and assembly; gather feedback and refine tasks.
Q4 2026	Scale-up	Increase production rate; target 1,000 units by year-end.
2027	Consumer Pilot	Launch early consumer pilot for limited household tasks; gather user feedback and adjust design.
2028-2029	Mass Market Rollout	Reach tens of thousands of units annually; expand use cases to retail, hospitality, and healthcare.

1.9 9. Stakeholders and Responsibilities

- **Executive Sponsor:** Elon Musk – Overall vision, resource allocation, and public commitment.
- **Product Manager:** Responsible for defining requirements, coordinating engineering and marketing.
- **Engineering Lead:** Oversees mechanical, electrical, and software development; ensures integration of tendon-driven hand and AI control.
- **Manufacturing Lead:** Manages supply chain, tooling, and assembly lines; coordinates with Fremont and Texas Gigafactories.
- **Safety & Compliance:** Ensures adherence to emerging robot safety standards and addresses liability issues.
- **AI/ML Team:** Develops neural network models and simulation infrastructure; maintains training data pipelines.
- **External Partners:** Component suppliers (Sanhua, Green Harmonic, Tuopu, Keli Sensing); regulatory agencies for compliance.

1.10 10. Future Considerations

- **Improved Power Systems:** Explore advanced battery chemistries (e.g., solid-state) to extend runtime beyond 10 hours.
- **Enhanced Sensing:** Incorporate haptic feedback, temperature sensors, and better visual mapping to enable more complex tasks.
- **Human-Robot Interaction:** Develop natural language capabilities and more expressive faces to improve interaction with humans.
- **Remote Software Updates:** Implement secure over-the-air updates for continuous improvement and bug fixes.
- **Robot-as-a-Service Model:** Consider subscription or leasing models to reduce upfront cost and drive adoption.

1.11 11. Conclusion

Tesla's Optimus Gen3 represents an ambitious leap toward general-purpose humanoid robotics. By incorporating a 22-DoF tendon-driven hand and focusing on mass production, the company aims to create a robot capable of performing industrial and household tasks while remaining affordable. However, significant engineering, supply chain, and economic challenges remain, particularly around dexterity, battery life, and production scaling. Success will depend on overcoming these obstacles, maintaining supplier relationships, and delivering demonstrable value in real-world use cases.